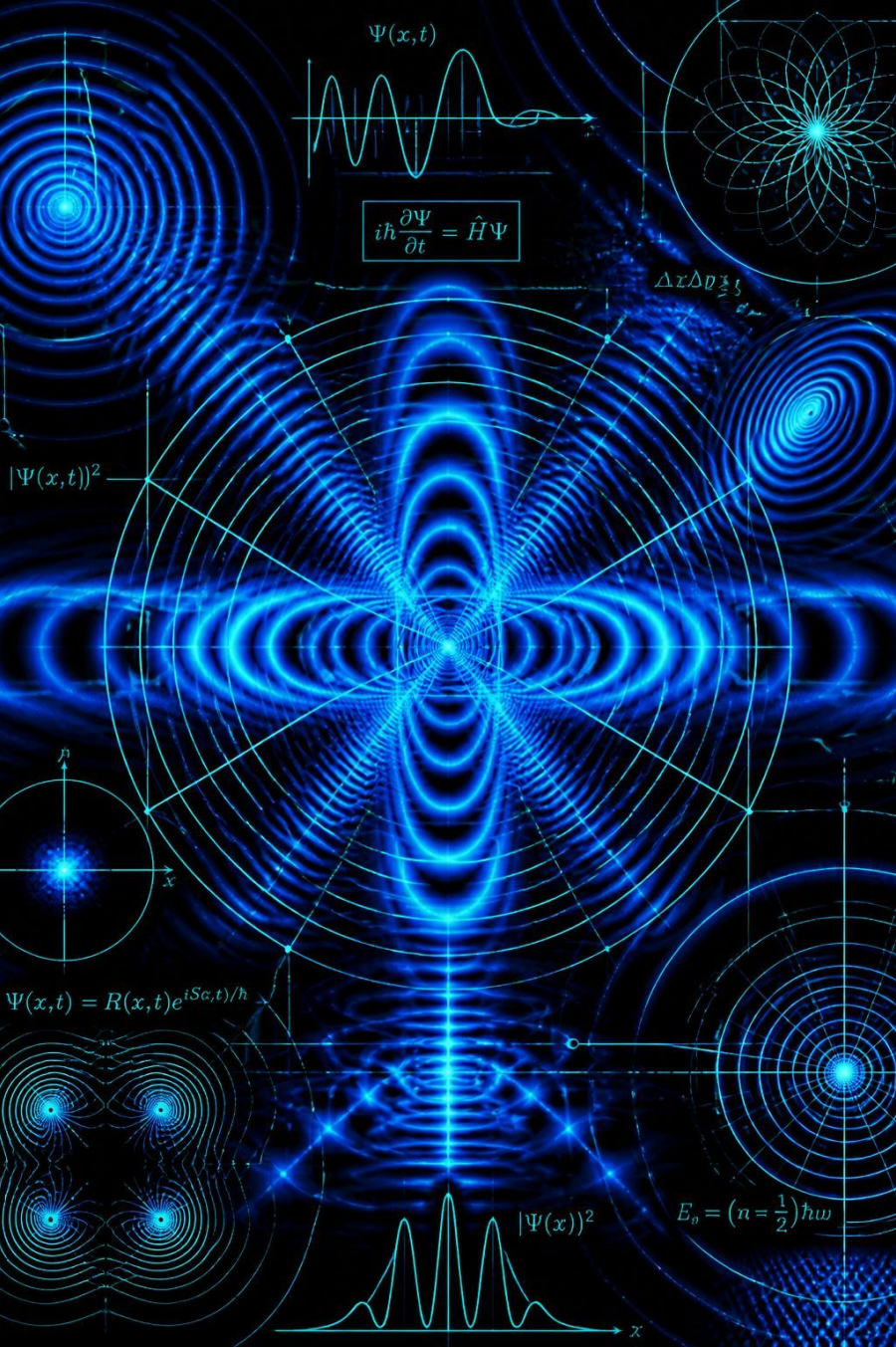




Dirac Notation and State Vectors

A mathematical language for the quantum world — compact, elegant, and powerful.

QUANTUM COMPUTING

What Is Dirac Notation?

Introduced by **Paul Dirac** in the 1930s, Bra-Ket notation is the standard mathematical shorthand of quantum mechanics and quantum computing. It encodes quantum states, operators, and measurements in a form that is both compact and computationally tractable.

Compact

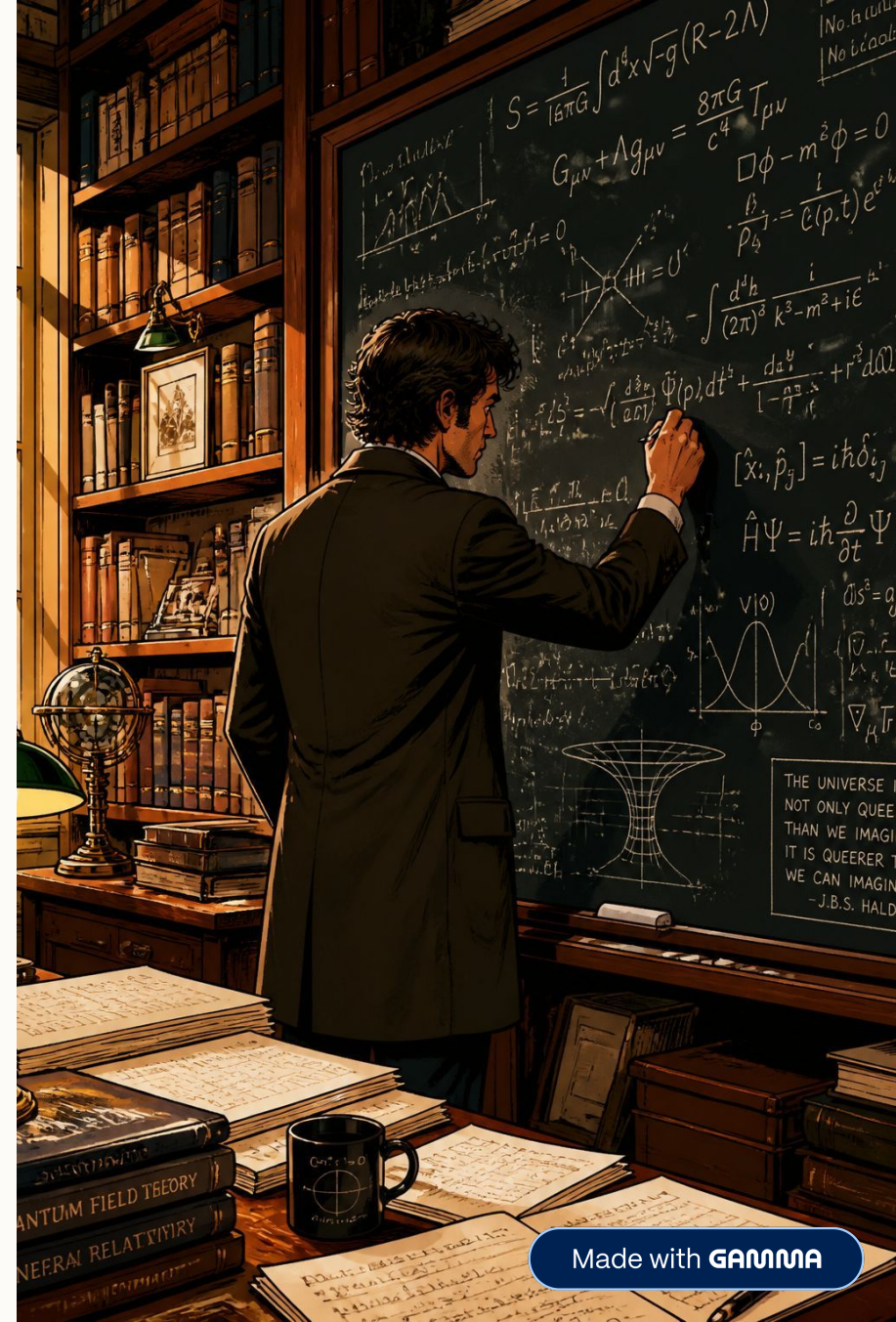
Replaces bulky matrix expressions with clean symbolic forms

Universal

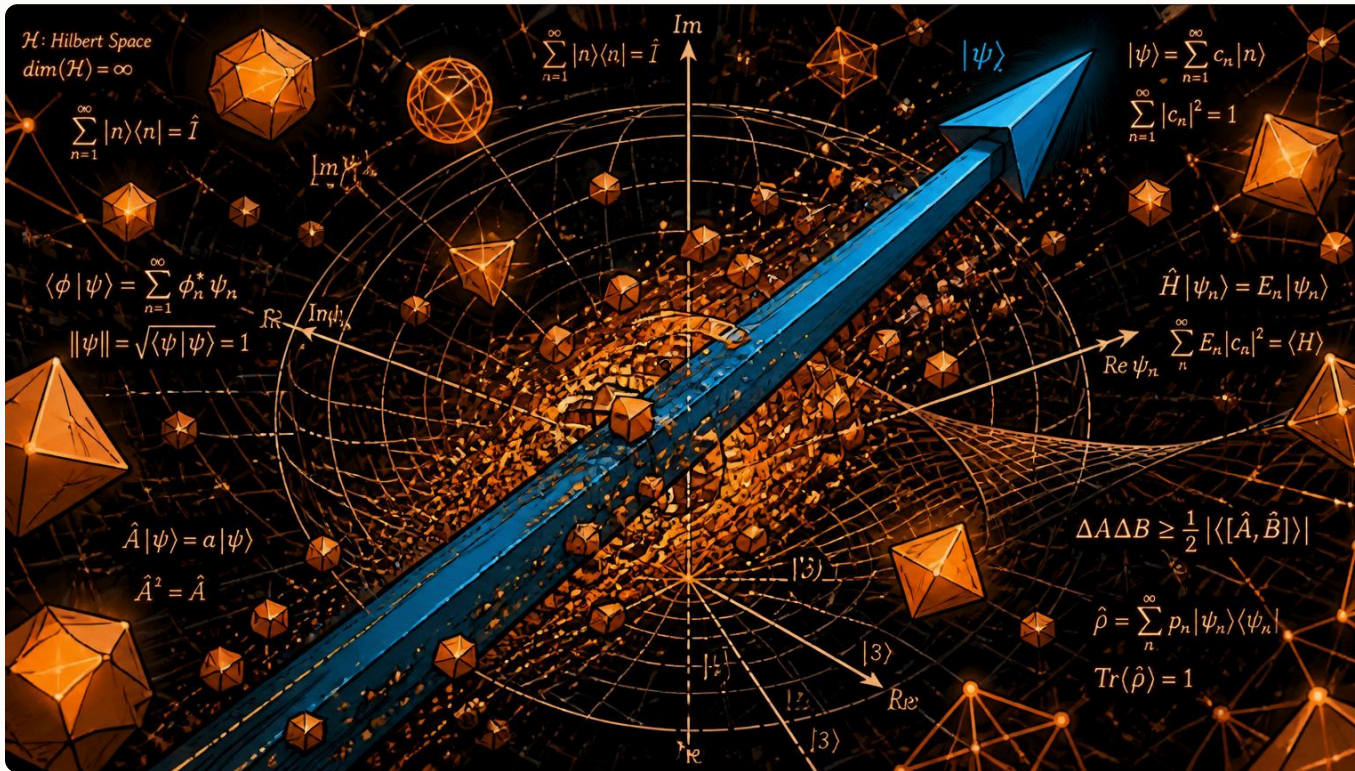
The lingua franca of quantum algorithms and circuit design

Efficient

Simplifies operations on qubits and quantum gates



State Vectors: Describing Quantum States



A **state vector** captures the complete physical state of a quantum system. It is written as a **Ket** $|\psi\rangle$ and lives in a complex vector space called a Hilbert space.

The two **computational basis states** of a qubit are:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

These correspond to the classical bit values 0 and 1 — but a qubit can be far more than just one or the other.

Bra and Ket: Two Sides of the Same Coin

Ket — Column Vector

Represents a quantum state as a column vector in Hilbert space.

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Bra — Conjugate Transpose

The Bra $\langle\psi|$ is the Hermitian conjugate of the Ket — a row vector with complex-conjugated entries.

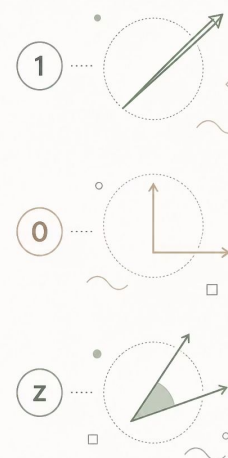
$$\langle\psi| = [\alpha^* \quad \beta^*]$$

$$\langle 0| = [1 \quad 0]$$

The Inner Product: Measuring Overlap

The **inner product** $\langle\phi|\psi\rangle$ quantifies how much two quantum states overlap — a complex scalar that encodes similarity and measurement probabilities.

❏ Orthogonal states are perfectly distinguishable — measuring one will never yield the other.



Two quantum states and their inner product outcomes:

when states are identical the result is 1 (perfect overlap)

when states are orthogonal the result is 0 (no overlap)

for general states the result is a complex number between 0 and 1 representing partial overlap

$$\langle 0|0\rangle = 1$$

Identical states — perfect overlap

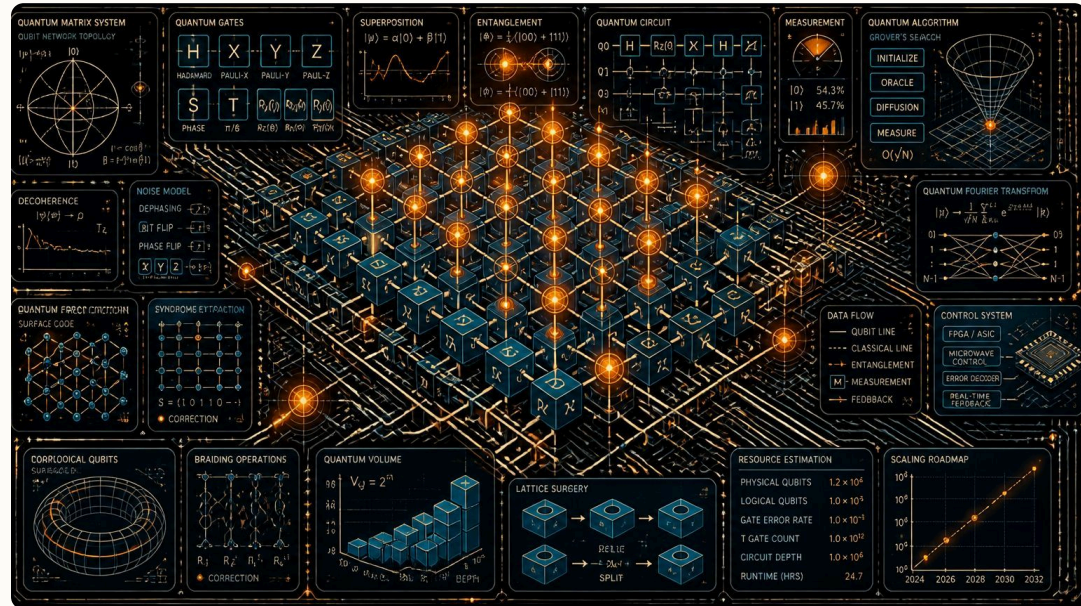
$$\langle 0|1\rangle = 0$$

Orthogonal states — no overlap

The Outer Product: Building Operators

The **outer product** $|\psi\rangle\langle\phi|$ produces a matrix — a linear operator acting on the Hilbert space. It is the building block of quantum gates, projectors, and density matrices.

$$|0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$



Projection Operators

Isolate a specific basis component

Quantum Gates

Unitary transformations on qubits

Density Matrices

Describe mixed quantum states

Superposition: More Than One State at Once

A qubit can exist in a **linear combination** of basis states simultaneously. The general form is:

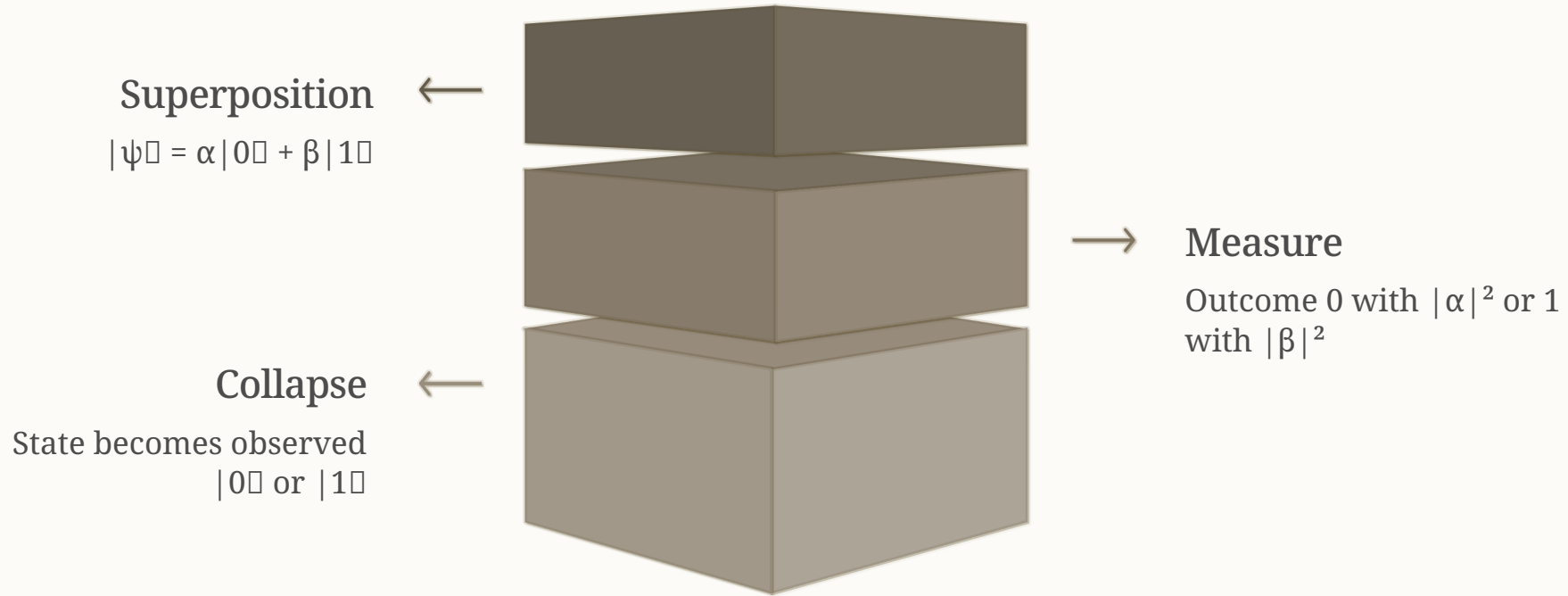
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Here α and β are complex **probability amplitudes** satisfying the normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1$$

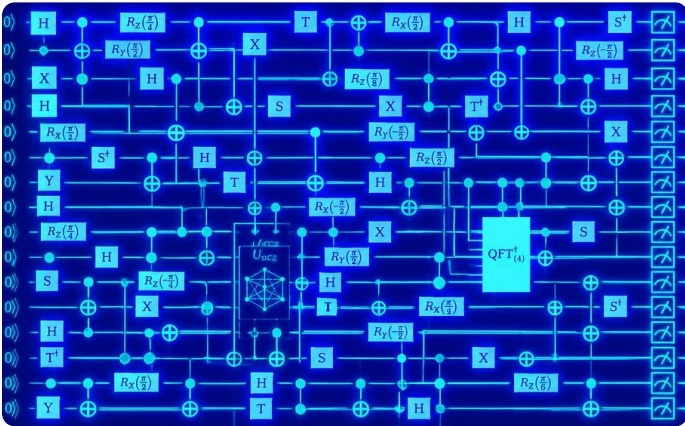
□ Example: the equal superposition state $|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$ gives a 50/50 chance of measuring 0 or 1.

Measurement: Collapse of the State



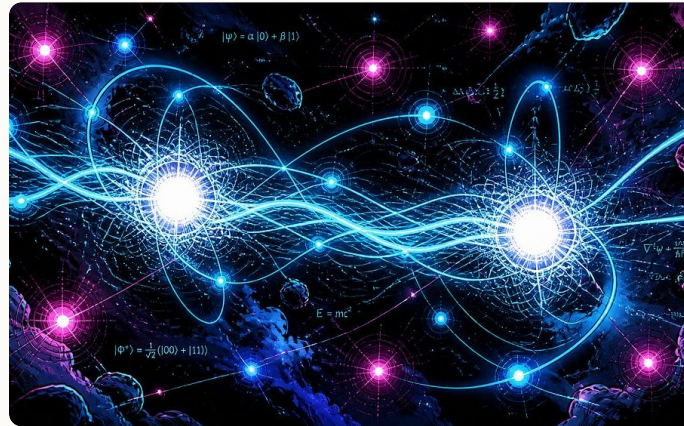
Measurement is inherently probabilistic and irreversible. Once a result is observed, the superposition is destroyed and the system collapses to the corresponding basis state — a fundamental departure from classical physics.

Where Dirac Notation Powers Real Applications



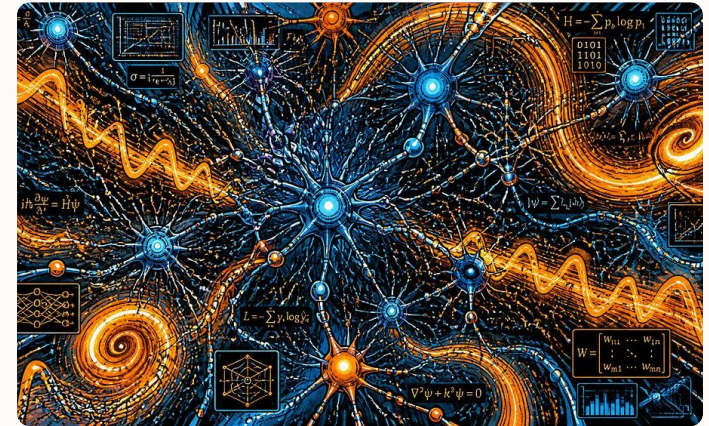
Quantum Circuits & Algorithms

Expressing gate sequences and algorithms like Shor's and Grover's



Teleportation & Cryptography

Describing entanglement, Bell states, and QKD protocols



Quantum ML & Error Correction

Encoding quantum data and protecting information from decoherence

Key Takeaways

01

Bra $\langle |$ and Ket $| \rangle$

Row and column vectors forming the backbone of quantum notation

03

Inner & Outer Products

Measure state overlap and construct operators, respectively

02

State Vectors

$|\psi\rangle$ fully describes a quantum system in Hilbert space

04

Superposition & Measurement

Qubits exist in multiple states until measurement collapses them probabilistically

- ❏ Dirac notation is not just shorthand — it is the conceptual framework that makes quantum computing readable, teachable, and computable.